INORGANIC TRENDS

GROUP – I (Alkali Metals)

#1) PROPERTIES

$Li > Na > K > Rb > Cs$

- → IE
- → EN
- → Metallic Bond strength.
- →

$Li < Na < K < Rb < Cs$

- → atm radii
- → Metallic character

⚠ $Li < K < Na < Rb < Cs$ } DENSITY

⚠ $Li > Cs > Rb > K > Na$ } REDUCING PROPERTY

#2) REACTIVITY

$Li < Na < K < Rb < Cs$ {General Reactivity}

$Li > Na > K > Rb > Cs$ {Complex forming Tendency}.

#3) PROPERTIES OF COMPOUNDS

(a) HYDRIDES

$LiH > NaH > KH > RbH > CsH$ (Stability)

$LiH < NaH < KH < RbH < CsH$ { → Ionic char^c, → Reducing nature }

(b) HYDROXIDES

$LiOH > NaOH > KOH > RbOH > CsOH$ {lattice enth}

$LiOH < NaOH < KOH < RbOH < CsOH$ { → Basic strength, → Solubility }

(c) HALIDES

$(X = Cl, Br, I)$

$LiF > NaF > KF > RbF > CsF$ $\{-\Delta_f H\}$

$LiX < NaX < KX < RbX < CsX$ $\{-\Delta_f H\}$

$LiF < NaF < KF < RbF < CsF$

$LiBr > NaBr > KBr > RbBr > CsBr$

$NaI > LiI > KI > RbI > CsI$

$CsCl > RbCl > LiCl > NaCl > KCl$

} Solubility

* For carbonates / Bicarbonates / Nitrates / Sulphates

→ TD order ; Li salts have max TD tendency ; polarizing power

→ Solubility ; except halides solubility always ↑ from (Li → Cs)

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Group II (Alkaline Earth metal)

$\boxed{Be > Mg > Ca > Sr > Ba > Ra}$

- → IE
- → EN
- → Metallic bond st

$\boxed{Be < Mg < Ca < Sr < Ba < Ra}$

- → at radii
- → Metallic char^{r}

General Reactivity

$\boxed{Be < Mg < Ca < Sr < Ba}$

Properties of compounds

(a) Hallides :

$BeH_2 > MgH_2 > CaH_2 > SrH_2 > BaH_2$ (Stability)

$BeH_2 < MgH_2 < CaH_2 < SrH_2 < BaH_2$

- → Ionic char
- → Reducing p

(b) Hydroxides :

$Be(OH)_2 > Mg(OH)_2 > Ca(OH)_2 > Sr(OH)_2 > Ba(OH)_2$ (Lattic

$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

- → Solubil
- → Basic Strenc

(c) Hallides :

$BeX_2 < MgX_2 < CaX_2 < SrX_2 < BaX_2$ (Ionic)

$BeX_2 > MgX_2 > CaX_2 > SrX_2 > BaX_2$ (Tendency to hydroly

* For carbonates / bicarbonates / nitrates / sulphates

*

→ TD order ; Be salts always have max TD tendency ; polarizin

→ Solubility ; Use lattice enthalpy ;
- small anion & small calca
- large anion & large calca

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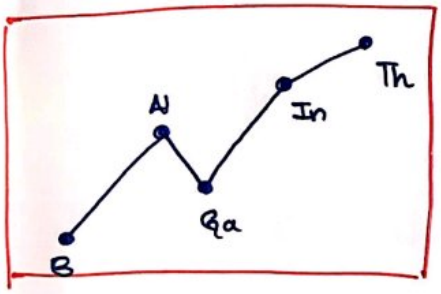
Group 13

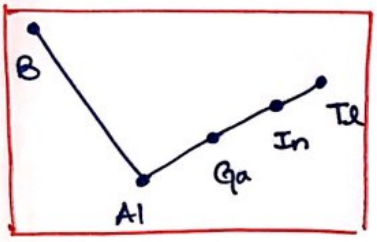
$B < Al < Ga < In < Tl$ { Density }



At radii

$B < Ga < Al < In < Th$

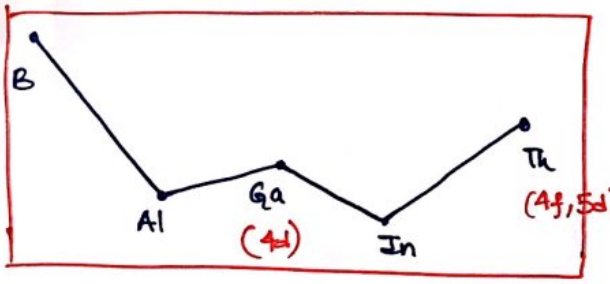
(Gallin has fully filled d orbital)



EN

$B > Tl > In > Ga > Al.$

Al → Tl less shielding



IE

$B > Tl > Ga > Al > In$

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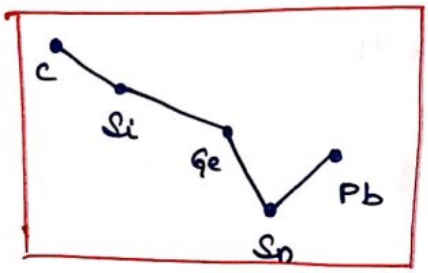
Group 14

$C < Si < Ge < Sn < Pb$

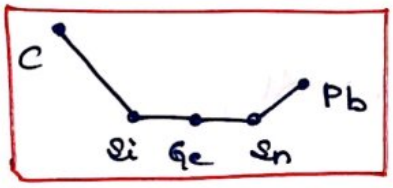
→ Covalent RADII
→ Metallic character



IE

$C > Si > Ge > Pb > Sn$

(unfo of $4f^{14}$ e^- in Pb)



EN

$C > Pb > Si \simeq Ge \simeq Sn$

(unf of $4f^{14}$ e^- in Pb)

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Group 15

$N < P < As < Sb < Bi$ — → At radii, → Density

$N > P > As > Sb > Bi$ — → IE, → EN

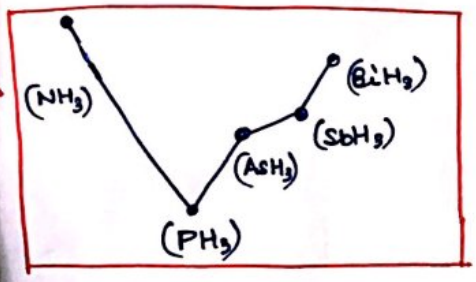
⚠ $P < N < As < Bi < Sb$ [Boiling Point]

Properties of compounds

→ **Hydrides**

$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ — → Bond angle, → Thermal stability, → Basic character

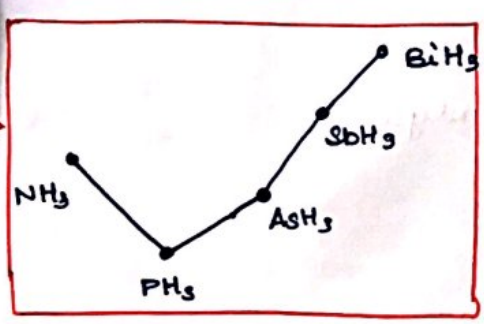
$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$ — → Reducing character



MP

$NH_3 > BiH_3 > SbH_3 > AsH_3 > PH_3$

(Vander Waals force not strong in solid)



BP

$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

(Vander Waals force strong in liq)

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Group 16

$O < S < Se < Te < Po$

$O > S > Se > Te > Po$

- → Atomic radii
- → Metallic character
- → IE
- → EN
- → Reactivity

⚠ $S > Se > Te > Po > O$ [Electron gain enthalpy]

Properties of compounds

Hydrides

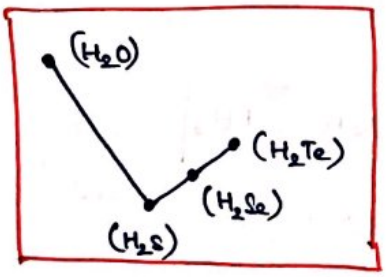
$H_2O > H_2S > H_2Se > H_2Te$

- → Bond angle
- → Thermal stability

$H_2O < H_2S < H_2Se < H_2Te$

- → Acidic strength
- → covalent character
- → Reducing nature

⚠



- → MP/BP
- → volatility

Halides

$SF_6 < SeF_6 < TeF_6$ {Order of hydrolysis}

SF_6 → exceptionally stable

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